Table 4: Demographic groups used in our steerability analysis.

Attribute	Demographic group
CREGION	Northeast, South
EDUCATION	College graduate/some postgrad, Less than high school
GENDER	Male, Female
POLIDEOLOGY	Liberal, Conservative, Moderate
INCOME	$100K+, <$30,000
POLPARTY	Democrat, Republican
RACE	Black, White, Asian, Hispanic
RELIG	Protestant, Jewish, Hindu, Atheist, Muslim

Table 5: LLMs we evaluate in our study. In some cases, we attempt to report size/training details of models to the best of our ability as these are often not clearly disclosed.

Model name	Provider	Size	Notes
j1-Grande	AI21 Labs	17B	Auto-regressive model from Lieber et al. (2021)
j1-Jumbo	AI21 Labs	178B	Auto-regressive model from Lieber et al. (2021)
j1-Grande v2 beta	AI21 Labs	17B	Instruct tuned version of j1-Grande, trained specifically to handle zero-shot prompts
ada	OpenAI	350M	Base GPT-3 model from Brown et al. (2020)
davinci	OpenAI	175B	Base GPT-3 model from Brown et al. (2020)
text-davinci-001	OpenAI	175B	Human-feedback model (Ouyang et al., 2022); trained via supervised fine-tuning on human-written demonstrations.
text-davinci-002	OpenAI	175B	Human-feedback model based on code-davinci-002 (Ouyang et al., 2022); trained via supervised fine-tuning on human-written demonstrations.
text-davinci-003	OpenAI	175B	Improved version of text-davinci-002 (Ouyang et al., 2022)

A.3 Models

For our analysis, we use a series of models from OpenAI and AI21 labs, detailed in 5. Since the model training process is not always publicly known, we attempt to report this to the best of our knowledge. Further documentation can be found `beta.openai.com/docs/model-index-for-researchers` and `docs.ai21.com/docs/`.

Once we prompt a model with a given question, we simply evaluate the log probabilities that each of the answer choices is the next-token. We then take these token log probabilities for each answer, exponentiate and then normalize them to get the model opinion distribution, i.e., $D_M = [e^{lp_A}, e^{lp_B}, ...,]/sum([e^{lp_A}, e^{lp_B}, ...,])$

Currently, OpenAI and AI21 limit the number of log probabilities they return via their API to 100 and 10 respectively. Thus, if one of the option choices (say 'A') is not in the set of returned log probabilities, we attempt to bound it as follows. Let's say the model returns a set of K (100 or 64) token-log probabilities pairs $\{t_k, lp_k\}$. We compute the total assigned mass as $p_{assigned} = \sum_{k \in K} e^{lp_k}$. The remaining mass is thus $p_{missing} = 1 - M$. We also find $p_m in = \min_{kinK} e^{lp_k}$, i.e., the minimum probability assigned to any of the K token choices. Then, we assigning the missing token 'A' the probability $min(p_{missing}, p_{min})$. Note that this is an upper bound on the true probability mass the model assigns to token 'A'.

As a baseline, we also consider a random model that chooses one of the answers choices per question at random.

A.4 Metrics

To compute the Wasserstein distance between human and LM opinion distributions to a question, we must map the to options to a metric space. To do so, we leverage the ordinal structure of the options (as provided by Pew surveys). For instance, we would map the set of options 'Strong Agree', 'Agree', 'Maybe', 'Disagree' and 'Strong Disagree' to the integers 1 through 5. We follow this approach in most cases, with the exception being questions for which the penultimate option is non-ordinal. For instance, if the choices were 'Very good', 'Very bad', and 'Neither good nor bad'. In this case, we map the answers to 1, 2 and 1.5 respectively.

A.5 Temperature scaling

In Section 4.1, we compare the model opinion distribution to a sharpened version of its human counterpart. This sharpening makes the human opinion distribution collapse towards its dominant mode. To do so, we use the standard temperature scaling approach from Guo et al. (2017). We use a temperature of 1e-3 in our analysis, but find that our results are fairly robust to the choice of temperature.